

MHSE27b: Applied Watershed Management - Course Assignment (2025)

**Hydro-Numeric Modelling (1D & 2D) and Analysis of the Mulde River at Auslaufbauwerk
Muldestausee section using HEC-RAS, Basement & QGIS**

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ASSIGNMENT

This assignment is about the utilization of 1D and 2D hydro-numeric (HN) models to analyse the flow characteristics of the Mulde floodplain during a flood event, focusing on areas prone to severe flooding for better management and understanding of the floodplain. The primary goal is to evaluate flood extent, water velocity, and discharge utilizing HEC-RAS for 1D modelling and QGIS BASE-mesh, Basement for 2D simulations. Moreover, the task involves expanding the BASEMENT model to investigate enhancements like parameter analyses, infrastructure modifications, or model calibrations and offers an additional chance to contrast the simulation results with the flood hazard maps created by state authorities and actual readings from prior flood occurrences, in which this report especially considers weirs at the specified locations and with varying discharge rates, that have been simulated. The simulation outcomes were to be assessed through both quantitative and qualitative methods, drawing on existing understanding of hydrology and hydrodynamics.

SUMMARY/ABSTRACT

This report presents a comprehensive hydraulic investigation of the Mulde River reach, utilizing a transition from 1D steady-flow modeling in HEC-RAS to 2D spatial analysis using QGIS and the BASEMENT plugin. The primary objective was to conduct and evaluate flood simulations to understand the hydraulic behavior of the reach under varying flow conditions. By comparing the traditional 1D approach with 2D spatial modeling, the study provides a detailed view of how complex topography and infrastructure influence flood propagation.

The study area was defined using a high-resolution Digital Elevation Model (DEM) and detailed river stationing, establishing a 1D baseline that identified critical water surface elevations and velocity distributions. For the 2D simulation, a computational mesh was developed with variable discretization based on land-use categories. This mesh was refined to include a 30 m² element area for the river channel to capture in-bank flow dynamics, while larger elements up to 1000 m² were used for urban areas to maintain computational efficiency. Roughness parameters were applied using spatially varying Strickler coefficients assigned to specific land-use regions to accurately parameterize flow resistance across the floodplain.

Simulation scenarios were conducted for steady-state discharges of 500 m³/s and 800 m³/s, incorporating hydraulic control structures including two weirs at Jeßnitz and one at Raghun. The results for the 500 m³/s scenario indicate that water levels remain largely contained within the existing dike system, with the weirs exerting a localized influence on the water surface profile without causing significant overtopping. In contrast, the 800 m³/s simulation demonstrated a significant increase in inundation extent. Under this extreme flow condition, the water spread extensively into agricultural and residential zones as the system's capacity was exceeded, highlighting the threshold at which the current infrastructure and floodplain geometry struggle to contain the discharge.

Ultimately, these simulations provide a detailed technical basis for comparing 1D and 2D modeling approaches. The findings emphasize the importance of spatial analysis in identifying areas of high velocity and depth, offering valuable insights into the flood characteristics and hydraulic performance of the Mulde River floodplain under different stress levels.

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USED ABBREVIATIONS

1D	One-Dimensional
2D	Two-Dimensional
HN	Hydro-Numeric
GIS	Geographic Information System
BASEMENT	Basic Simulation Environment
TIFF	Tag Image File Format
CUM	Cubic Meters per Second
CRS	Coordinate Reference System
DEM	Digital Elevation Model
ETRS	European Terrestrial Reference System
M.A.S.L.	Meters Above Sea Level
OSM	Open Street Maps

1. MOTIVATION AND OBJECTIVE

1.1 MOTIVATION

The Mulde River is a major tributary of the Elbe and is recognized as one of the most dynamic and flood-prone river systems in Central Europe. Historically, flooding events have caused severe economic and environmental damage, with a recorded peak discharge of 2,010 m³/s, before modern flood management systems were fully implemented (*Supplement Material*).

The "Complexity of Flow" in this region presents a significant engineering challenge. Under normal conditions, water is confined to the main channel; however, during extreme events, the river interacts dynamically with its floodplains, dense riparian vegetation, and man-made structures such as the weirs at Jeßnitz and Raghun. Vegetation increases hydraulic roughness (Chow, 1959). Furthermore, structures like weirs create backwater effects that can lead to localized dike overtopping.

Because physical measurements during such high-velocity flood events are often dangerous or technically impossible to conduct in the field, Hydro-Numeric (HN) models serve as the primary tool for modern flood management. These models allow engineers to simulate "what-if" scenarios safely, providing the necessary data to plan protection measures—such as dikes or the strategic placement of infrastructure—without risking human life or equipment during actual disasters.

1.2 OBJECTIVES

The study aims to achieve the following objectives:

- I. To develop a 1D HEC-RAS model to provide an initial hydrodynamic assessment of the Mulde floodplain and analyze the flood area and flow characteristics of study area.
- II. To construct a 2D BASEMENT model in QGIS BASE mesh and analyze detailed flood extent, water depth, water velocity distribution and terrain interaction during a flood event.
- III. To evaluate different discharge scenarios ($Q = 500$ and $800 \text{ m}^3/\text{s}$) to study the impact of increasing flood magnitudes on floodplain dynamics, water depth and flow velocities.
- IV. To assess the influence of weir structures on flooding behavior by incorporating inner boundary conditions for Jeßnitz weirs and Raghun weir.

2. METHODOLOGY

This section outlines the systematic approach used to simulate the flood dynamics of the Mulde River floodplain. It describes the integration of geospatial data and hydraulic modeling tools to evaluate the impact of land cover and infrastructure on water flow.

2.1 STUDY AREA

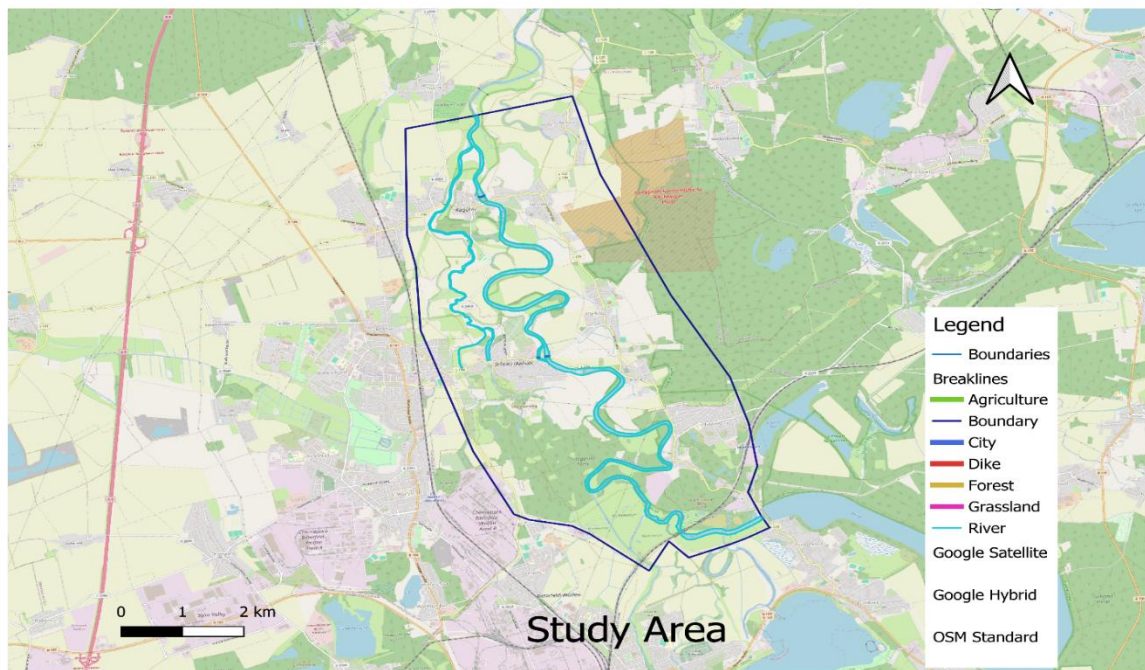


Figure 1: Study Area of the Mulde River floodplain

The study area is situated within the Mulde River floodplain in Sachsen-Anhalt, Germany, focusing on the critical reach extending from the Muldestausee reservoir's outlet structure (Auslaufbauwerk) through the downstream floodplain. This region is a vital component of the local flood protection strategy, characterized by a complex hydraulic system regulated by key structures such as the weirs at Jeßnitz and Raghun. To protect surrounding communities and small settlements like Jeßnitz, the river is flanked by dikes ranging from 2.5 to 5 meters in height. The area features a diverse landscape of agricultural fields, grasslands, forests, and urban areas, all of which contribute to varying surface roughness and influence flow resistance, a concept fundamentally detailed in engineering by *Chow (1959)*. With a design discharge capacity of approximately $1,200 \text{ m}^3/\text{s}$, the management of this area is essential for resilience against periodic extreme events which saw peak discharges reach $2,010 \text{ m}^3/\text{s}$. Consequently, the site necessitates continuous risk management and high-fidelity Hydro-Numeric (HN) modeling to safeguard infrastructure and local populations.

2.2 DATA ACQUISITION

- I. The Digital Elevation Model (DEM) was obtained from the course instructor to define the terrain of the Mulde floodplain.
- II. The EPSG:25832 projection file was downloaded from spatialreference.org to

ensure the model was correctly georeferenced.

- III. Land cover classification was determined in QGIS following the specific instructions provided in the assignment.
- IV. The study area was categorized into agricultural fields, grasslands, forests, and urban centers to analyze surface characteristics.
- V. Strickler values were assigned to these land cover classes to accurately parameterize surface roughness and flow resistance.
- VI. All data was made ready for sequential processing in HEC-RAS for 1D assessment and BASEMENT for 2D hydrodynamic simulation.
- VII. Final input parameters and boundary conditions were implemented step-by-step to complete the model setup for the Mulde River system.

2.3 1D-HN MODEL

The HEC-RAS 1D model was developed to analyze the longitudinal hydraulic characteristics and discharge distribution of the Mulde floodplain. This assessment specifically focuses on evaluating water surface profiles and flow velocities across the study area under defined flood conditions.

2.3.1 HEC-RAS 1D Model Setup and Simulation

- I. The modeling process began with the integration of the terrain file (DEM) to provide a precise elevation foundation for the Mulde River floodplain.
- II. The river geometry was then constructed by digitizing the centerline, side lines, and bank lines to define the primary flow paths.
- III. A total of 34 cross-sections were drawn perpendicular to the river flow to capture the detailed physical characteristics of the channel and surrounding floodplain.
- IV. Hydraulic parameters were assigned to the model, including a bed slope of 0.0005 and specific surface roughness values represented by Strickler coefficients of 30 and 25.
- V. A steady flow condition with a baseline discharge of 62 m³/s was applied as an upstream boundary condition to simulate the initial flood scenario.
- VI. The model was computed and analyzed within the HEC-RAS results section to evaluate longitudinal water levels and flow velocities across the study area.

2.4 2D-HN MODEL

The 2D hydraulic model was developed using the BASEMENT plugin within QGIS to conduct a high-resolution spatial analysis of the Mulde floodplain dynamics. BASEMENT is a specialized simulation tool developed by the Laboratory of Hydraulics, Hydrology, and Glaciology (VAW) at ETH Zürich to model hydro- and morphodynamics. For this study,

version 4 was utilized, which performs two-dimensional simulations based on the Saint-Venant equations. This software provides a robust framework for capturing complex flow interactions across the Mulde floodplain. Unlike 1D models that provide only longitudinal flow characteristics, this 2D approach captures complex lateral flow interactions, spatial velocity distributions, and detailed flood depths across the entire model domain. A computational mesh was generated based on the provided DEM, and the previously determined land cover classifications were used to apply spatially varying Strickler roughness values. The model was configured to simulate both a baseline discharge of 500 m³/s with and without weir and 800 m³/s with weir and providing a comprehensive understanding of flood behavior and potential risks within the dikes.

2.4.1 MESH CREATION IN QGIS

- I. The project Coordinate Reference System was set to ETRS89 / UTM zone 32N (EPSG:25832) to ensure spatial accuracy.
- II. The provided Digital Elevation Model (DEM) and shapefiles were imported as the topographic foundation for the study area.
- III. An OSM Google Satellite layer was added to visualize the study area and assist in the accurate placement of model components.
- IV. Model boundaries were defined by identifying the INLET (located after the Pretziener Weir) and the OUTLET.
- V. River cross-sections were delineated and model breaklines were constructed to divide the domain into distinct hydraulic regions based on land use and dike locations.
- VI. A Quality Mesh was generated using the BASEMesh plugin, incorporating the defined boundaries, breaklines, and land-use regions.
- VII. Elevation data from the DEM was interpolated onto the quality mesh to create the final computational mesh required for the simulation.
- VIII. The model was executed in BASEMENT to capture the spatial distribution of water depth and velocity across the Mulde floodplain.

Table 1:2D Mesh Discretization Parameters by Land Use Region

Region	Material ID	Maximum Element Area [m ²]
River	1	30
Agriculture	2	500
Forest	3	250
Grassland	4	250
City	5	1000
Dike	6	10
Hole	8	50

2.4.2 MODEL SETUP IN BASEMENT

Following the initial setup in QGIS, the model was configured in BASEMENT through the following geometry and hydraulic input stages:

- I. Geometry and Region Definition
 - a) The computational mesh generated in QGIS was loaded as the primary geometric foundation.
 - b) An interpolation layer was added using the "weighted" method to refine topographic representation.
 - c) Model regions and strings (INLET, OUTLET, Weir_Inlet, Weir_Outlet) were defined to match the QGIS spatial boundaries, with the upstream direction set to "right" for all strings.
- II. Hydraulic Boundary Conditions
 - a) Standard Boundaries: The inlet was set as a uniform inflow (uniform_in) and the outlet as a uniform outflow (uniform_out), both utilizing a slope of 0.005%.
 - b) Discharge Scenarios: Simulations were conducted using inflow discharges of 500 m³/s and 800 m³/s across different scenarios.
 - c) Linked Boundaries (Weir): A weir condition was implemented using a 0.7 Poleni factor and an crest elevation of 73 m; this height was selected to prevent backwater effects and potential city inundation, as the surrounding DEM elevations range between 74 m and 75 m.
- III. Friction and Surface Roughness Roughness coefficients were assigned to the defined regions using Strickler (Kst) values to simulate flow resistance classified as on the table 2.
- IV. Model Parameters and Simulation Settings
 - a) Initial Conditions: The model was initialized in a "dry" state.
 - b) Numerical Parameters: The CFL was set to 0.7, the maximum time step to 100 seconds, and the minimum water depth to 0.005 m.
 - c) Physical Properties: Default values were maintained, including a fluid density of 1000 kg/m³ and gravitational acceleration (g) of 9.81 m/s².
 - d) Safety: Safe Mode was enabled to ensure simulation stability

A comprehensive record of the detailed setup and full simulation data is attached in the Appendix section of this report.

Table 2: Roughness Coefficients *K_{st}* for Land Use Categories

Region	Material ID	Strickler Coefficient, $K_{st} [(m^{1/3})/s]$
River	1	30
Agriculture	2	20
Forest	3	8
Grassland	4	20
City	5	35
Dike	6	35

2.5 MODEL SIMULATION

The final modeling phase utilized the Simulation feature within BASEMENT to compute hydraulic variables based on the previously configured setup file. The model was configured to calculate and export water depth, flow velocity, friction (Chezy), specific discharge, and water surface elevation (WSE). To ensure the simulation reached a steady-state condition, a total duration of 172,800 seconds (48 hours) was selected, with results recorded at intervals of every 1,800 seconds. Numerical stability was maintained by implementing an initial time step of 0.002 seconds and a minimum time step of 0.0001 seconds, ultimately generating the comprehensive data files required for spatial analysis.

2.6 MODEL OUTPUT

Upon completion of the simulation, data for water depth, flow velocity, friction, and specific discharge were obtained for all mesh grid points across the entire temporal domain. The results were exported from BASEMENT in .xdmf format and integrated with the computational mesh source in QGIS for spatial visualization. Using the Crayfish plugin, time-series data for flow characteristics at the inlet and outlet, along with water surface elevations for all cross-sections, were extracted as .csv files. These datasets were subsequently utilized to generate comparative plots in Excel, providing the basis for the detailed analysis and discussion presented in the following sections of this report.

3. ANALYSIS

The analysis was conducted in sequential phases, beginning with Task 1, which involved the development and hydraulic evaluation of the HEC-RAS 1D model to establish baseline flow characteristics. This was followed by the transition to 2D modeling using QGIS and BASEMENT, where a flood simulation for a steady-state discharge of 500 m³/s was performed to capture

spatial dynamics. For Task 2, the model was refined to include a weir structure, with simulations executed for both 500 m³/s and 800 m³/s discharge scenarios. These completed datasets now serve as the foundation for the comparative analysis of water levels, velocities, and inundation extents presented in the following sections.

3.1 HEC-RAS 1D HYDRAULIC ANALYSIS

The 1D simulation results were evaluated to establish a baseline for the river's longitudinal behavior and channel capacity. This analysis focused on the longitudinal water level profile to identify flow transitions and potential overtopping, while the energy grade line and critical depth were examined to determine prevailing flow regimes across different reaches. Furthermore, velocity distributions were analyzed at specific cross-sections to identify hydraulic bottlenecks, and the resulting flood inundation extent was mapped to visualize areas of containment and localized discrepancies between the 1D model and the digital terrain.

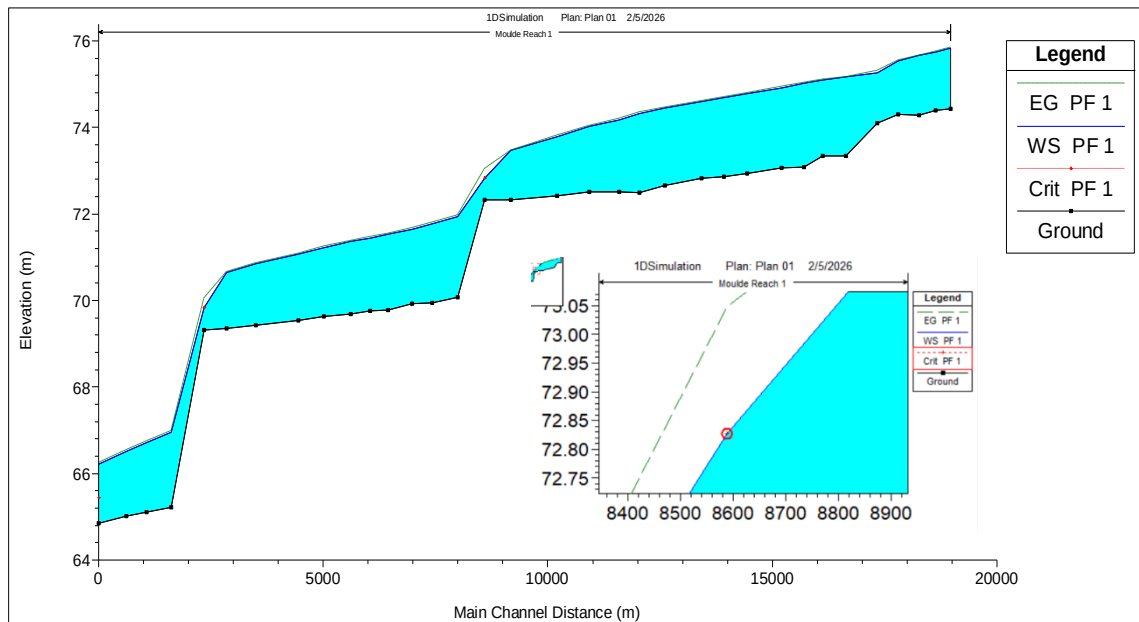


Figure 2: 1D Longitudinal Profile of the Mulde Reach

At a steady-state discharge of 62 m³/s, the Mulde Reach 1 predominantly exhibits subcritical flow conditions. However, the flow reaches critical depth at stations 2364, 8597, and 10, where a significant increase in the channel bed slope triggers localized high-velocity zones. While the average velocity across all 34 cross-sections is calculated at 0.75 m/s, a distinct spike in velocity is observed at these critical depth locations due to the steepened topographic gradient.

Table 3: Critical Hydraulic Summary Profile

Reach	River Station(m)	Profile	Water Surface Elevation(m)	Critical Water Surface Elevation(m)	Energy Grade Elevation(m)	Velocity (m/s)
1	8597	PF1	72.83	72.83	73.05	2.09
1	2364	PF1	69.84	69.84	70.05	2.04
1	10	PF1	66.22	65.45	66.25	0.86

Data derived from the summary table above highlights specific critical point stations where hydraulic transitions occur. A comparison shows that the 1D model often yields lower values than the QGIS spatial analysis. This discrepancy is due to the fixed placement of the 34 cross-sections; since sections were not positioned at every localized high and low terrain point, the 1D model lacks the resolution to capture the extreme values identified in the full QGIS dataset.

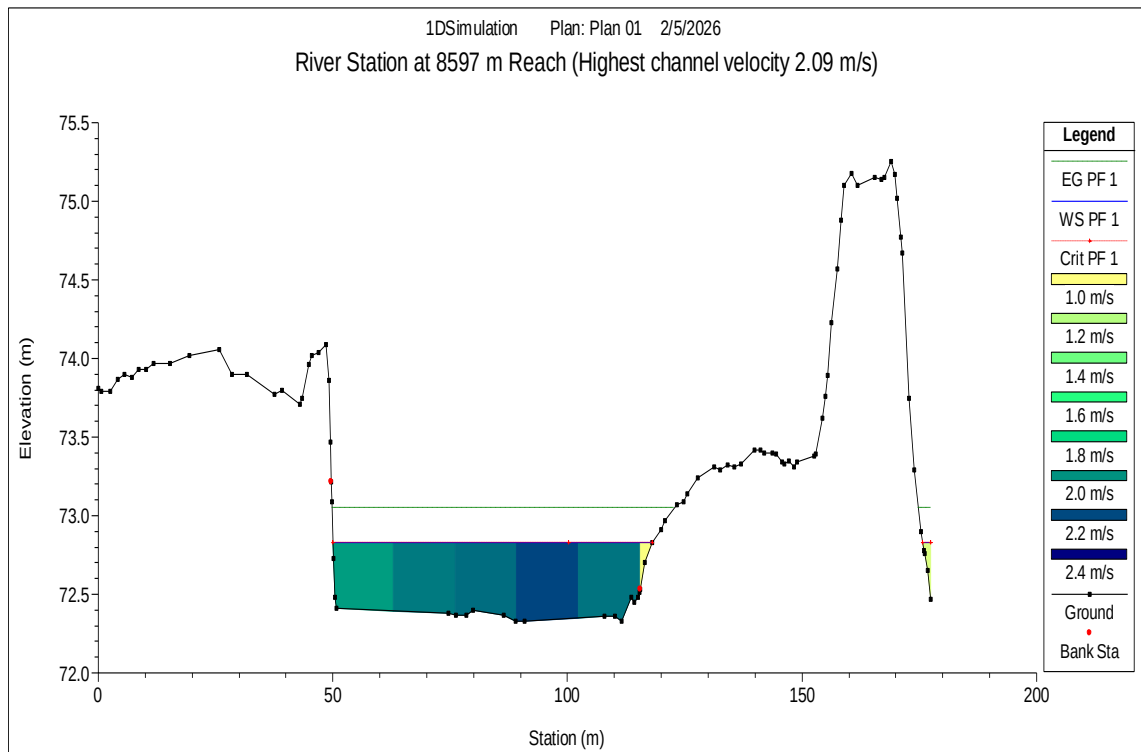


Figure 3: Cross-Sectional Velocity Distribution at River Station 8597 m

At this station, the flow has reached the Critical Depth. This is the specific point where the water is moving at the same speed as a wave traveling on its surface, creating maximum hydraulic efficiency but high instability. The steepness of the channel at this section provides the gravitational energy needed to accelerate the discharge of 62 m/s to its peak velocity of 2.09 m/s. To pass through this "pinch point," the river converts its potential energy (depth) into kinetic energy (speed). This results in a lower water level but a significant spike in velocity.

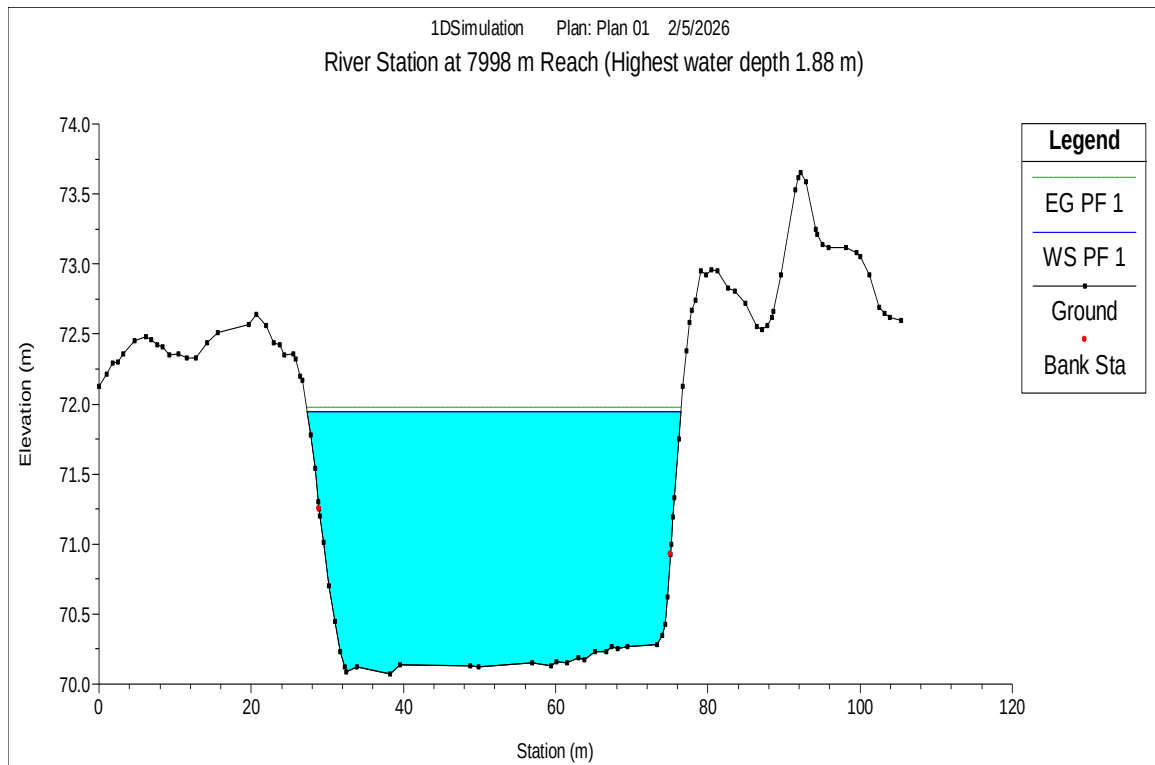


Figure 4: Cross-Sectional Geometry and Water Depth at River Station 7998 m

Section 7998 exhibits a higher water depth primarily due to a backwater effect caused by a massive reduction in energy slope. As the slope flattens from 0.011481 upstream (8597) to 0.000255 downstream, the water loses the gravitational momentum required to maintain high speeds, causing the velocity to drop from 2.09 m/s to 0.77 m/s. To maintain the constant flow rate of 62.00 m³/s, the water must "pile up," converting its kinetic energy into potential energy (depth). This results in a depth of 1.88 m at the downstream section compared to only 0.50 m upstream, as the flatter downstream reach acts as a hydraulic bottleneck.

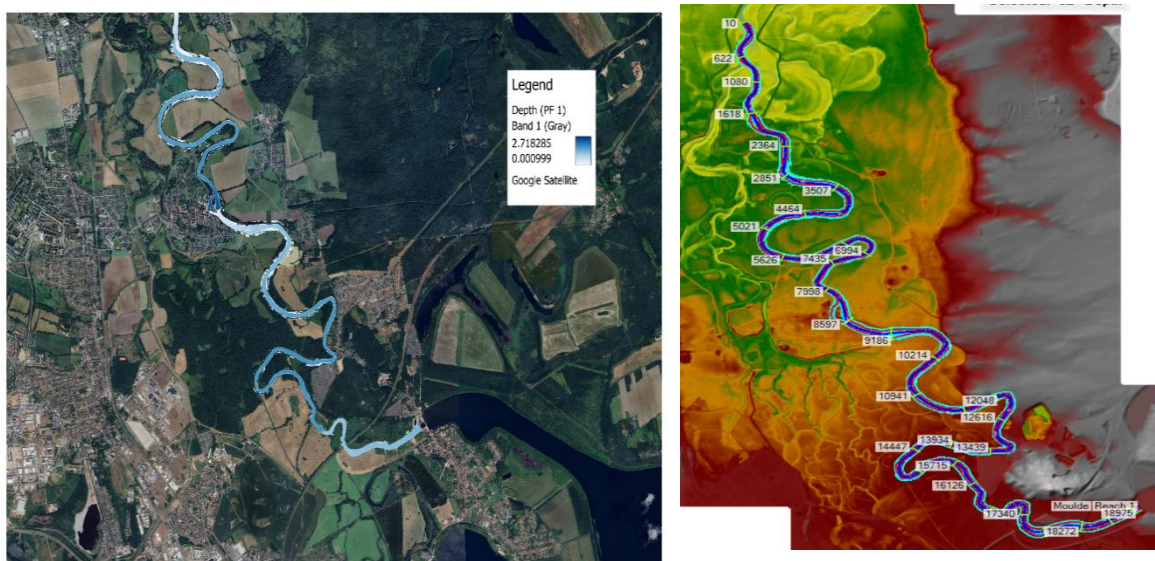


Figure 5: Inundation Depth Map over Google Satellite Imagery for the Mulde Reach and River Stationing and Digital Elevation Model (DEM) of the Study Area

The flood inundation mapping for the 62m³/sec discharge event reveals a high degree of channel containment, with no significant overbank flooding observed throughout the majority of the reach. However, localized discrepancies exist between the HEC-RAS 1D results and the QGIS spatial visualization. Specifically, field observations and GIS analysis indicate a localized depth of 2.43m immediately downstream of the weir, whereas the HEC-RAS model reports a maximum depth of 1.88m. This variance is primarily due to the spacing of cross-sections; since a section was not placed immediately at the weir's toe, the model cannot fully capture the localized 'scour hole' or the peak of the hydraulic jump. Furthermore, minor instances of water appearing outside the river boundary in the GIS map are attributed to the interpolation limitations of 1D modeling. In these areas, HEC-RAS linearly connects water surface elevations between distant cross-sections, occasionally 'bleeding' over the terrain's bank geometry where the actual channel might be narrower. Despite these mapping artifacts, the hydraulic logic remains consistent: high-velocity flow at Station 8597 m, 2.09m per sec transitions into a deep, subcritical backwater pool toward Station 7998.

3.2 QGIS AND BASEMENT 2D HYDRAULIC ANALYSIS (500 M³/S SCENARIO)

Following the 1D assessment, a two-dimensional simulation was conducted for a steady-state discharge of 500 m³/s using the BASEMENT software integrated with QGIS. This phase provides a detailed spatial analysis of the Mulde floodplain, specifically focusing on the distribution of water depths, flow velocities, and specific discharge across the mesh. Additionally, the analysis evaluates the Friction (Chezy) values to illustrate how varied surface roughness across regions like forests and riverbanks influences flow efficiency. These parameters collectively characterize the hydraulic behavior of the reach under high-flow conditions before the implementation of weir structures.

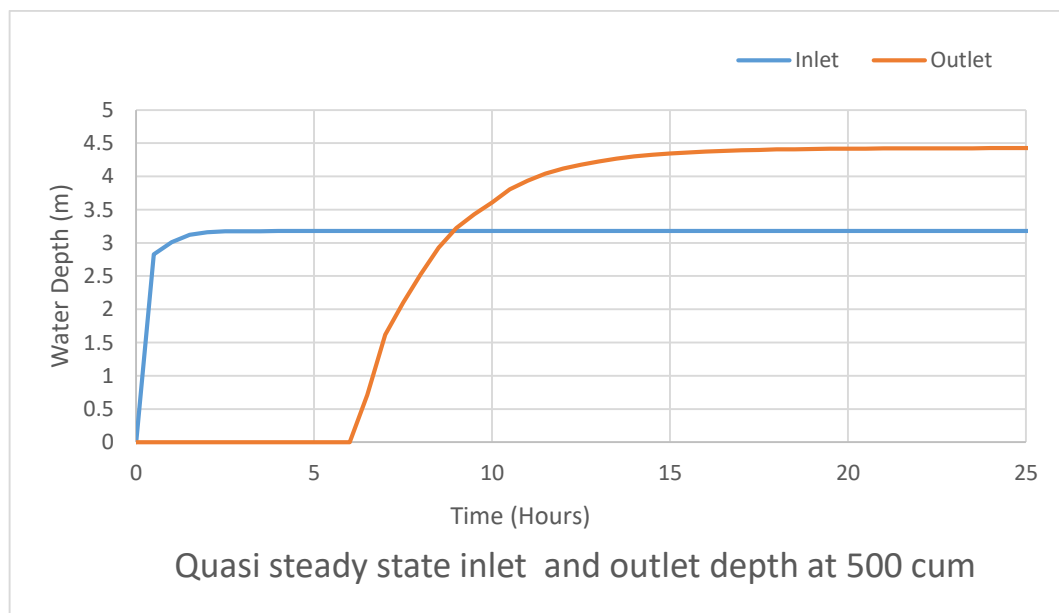


Figure 6: Hydrograph Comparison of Quasi-Steady State Inlet and Outlet Depths at 500 m³/s

The simulation transitions to a **quasi-steady state** at different intervals, with the inlet reaching a stable depth of **3.177 m** at 5 hours (18,000 s), while the outlet stabilizes at **4.425 m** after 25 hours (90000 s). Depth profiles show a clear longitudinal variation, with the highest overall depth of **4.77 m** occurring in the lower reach between the section of Raghun weir and the outlet.

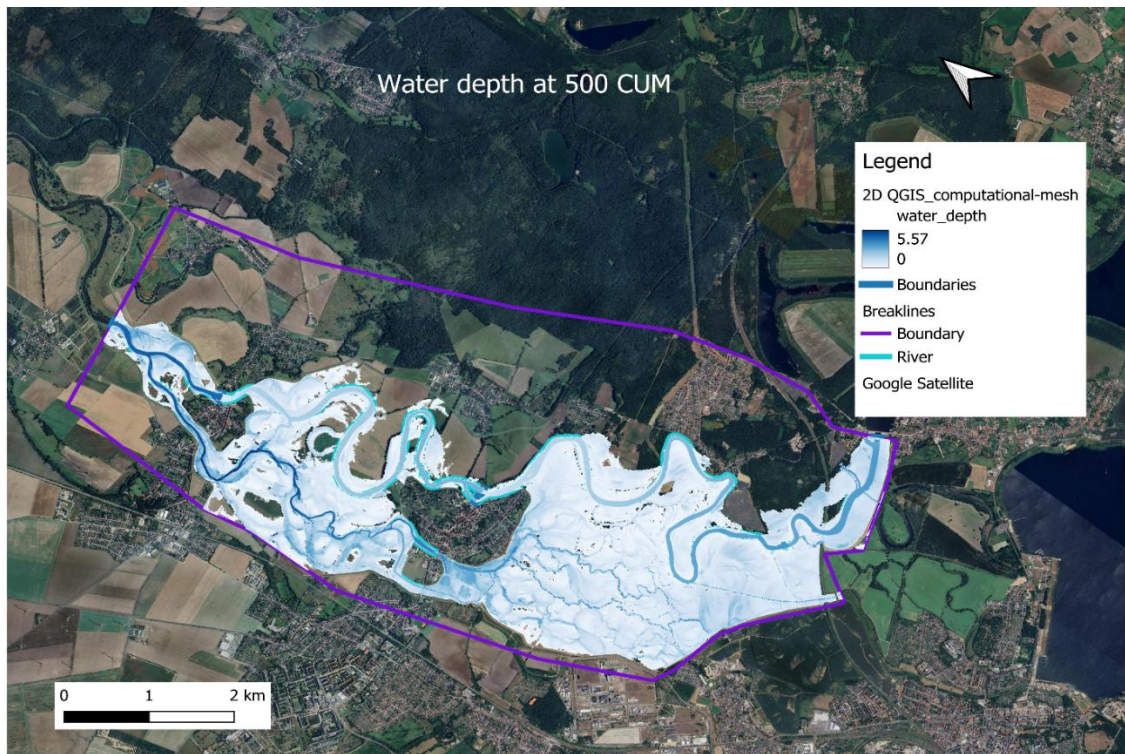


Figure 7: 2D Inundation Depth at 500 m³/s

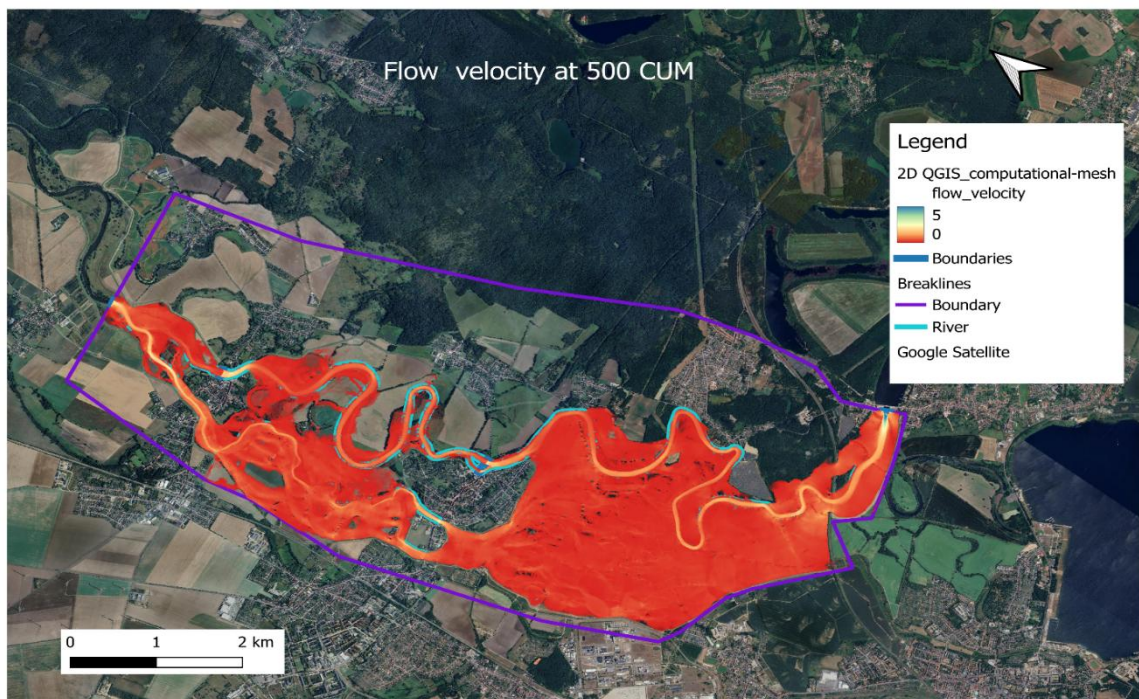


Figure 8: 2D Flow Velocity Map at 500 m³/s

Flow velocities fluctuate across the river's length, peaking at a maximum of 2.01 m/s just upstream of the Jeßnitz weir. Consistent velocity spikes are observed upstream of the proposed weir sections in Scenarios 2 and 3, as well as at the model inlet. The elevated entry velocity at inlet is likely due to the kinetic energy generated by an additional weir located immediately upstream of the computational boundary.

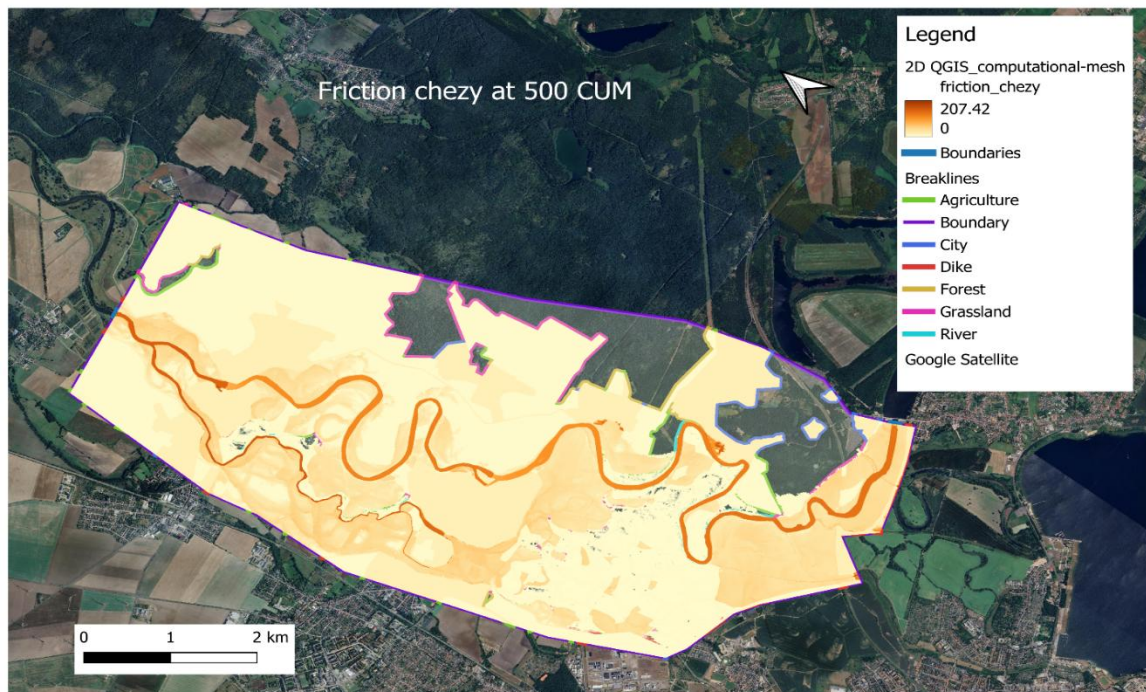


Figure 9: Spatial Distribution of Chézy Friction Coefficients for Land Use Categories

In this hydraulic model, the Chézy friction values represent the flow efficiency of the terrain rather than the physical magnitude of resistance. The River displays the highest values (115–155), as its smooth, unobstructed channel allows water to move with maximum velocity and minimal energy loss. Grassland and Agriculture provide moderate efficiency (45–70) due to surface vegetation, followed by Dikes and City areas (20–25). Conversely, the Forest exhibits negligible Chézy values, reflecting the highest physical roughness; the dense trees and undergrowth create massive resistance, which the model mathematically represents with a very low coefficient to "brake" the water flow. This ranking correctly illustrates that as physical obstacles increase, the Chézy value must decrease to accurately simulate slower, more turbulent movement across the landscape. The magnitude of specific discharge, measured in m^2/sec shows a localized peak at the model inlet compared to a stabilized value of upstream weir located outside the model boundary. Also the simulation results indicated that the specific discharge reached a value of approximately $9 \text{ m}^2/\text{s}$ at the outlet.

3.3 QGIS AND BASEMENT 2D HYDRAULIC ANALYSIS (500 M³/S SCENARIO WITH WEIRS)

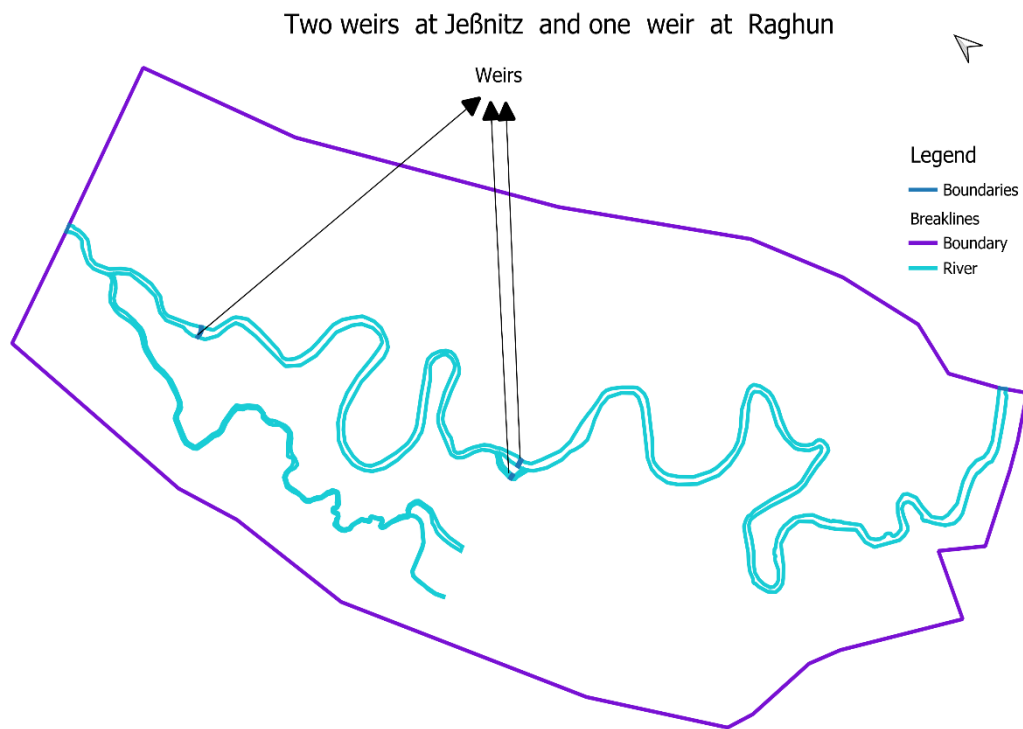


Figure 10: Locations of Hydraulic Control Structures (Weirs) at Jeßnitz and Raghun

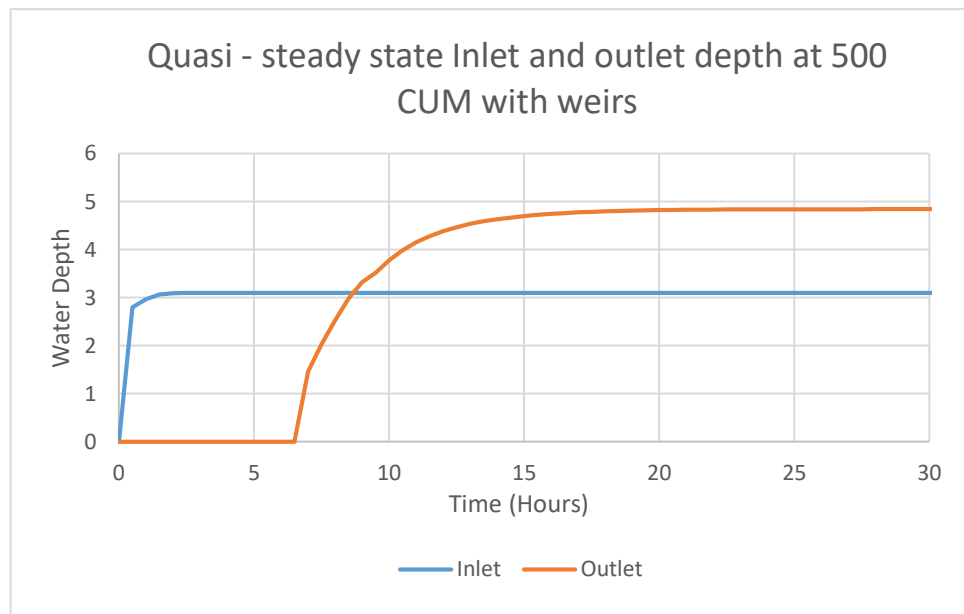


Figure 11: Quasi-steady State Hydrograph for Inlet and Outlet Depths with Weir Influence

The water depth reaches a quasi-steady state of approximately 3.097 m after about 6.5 hours (23,400 s), while the outlet depth reaches a steady value of 4.841 m after approximately 29.5 hours (106,200 s). Compared to the scenario without the weir, a slight

delay in reaching quasi-steady conditions is observed, along with a marginal increase in outlet water depth.

The simulation results indicate that the introduction of the weir does not cause any significant change in water depth. This is because the assumed weir crest elevation (73 m) at Basement remains approximately 20 cm below the water surface level under the 500 m³/s discharge condition, and the elevation difference between the weir section (≈ 72.4 m bed level) and the adjacent terrain (≈ 74 m urban area) checked from DEM is relatively small. Increasing the weir height further would likely induce backwater effects and pose a risk of urban inundation; therefore, the weir height was conservatively assumed. Under these conditions, the flow depth remains essentially unchanged.

No major changes are observed in flow velocity, specific discharge, or Chezy's friction coefficient. However, a very slight increase in inundation extent can be identified when analyzing detailed water-depth visualizations.

3.4 QGIS AND BASEMENT 2D HYDRAULIC ANALYSIS (800 M³/S SCENARIO WITH WEIRS)

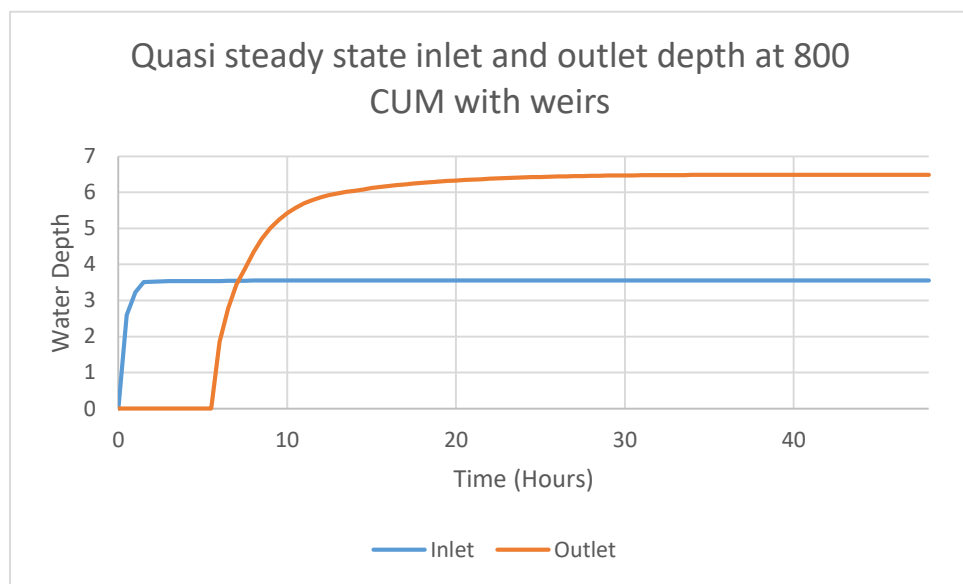


Figure 12: Quasi-steady State Hydrograph for Inlet and Outlet Depths at 800 m³/s with Weir Influence

In this scenario, the water depth increased globally across the entire river section, primarily driven by the high discharge of 800 m³/s. This increased volume overwhelms the channel's natural capacity, forcing a rise in stage from the inlet to the outlet. Superimposed on this global rise is the specific influence of the infrastructure; the section downstream of the Raghun weir shows the highest depths, reaching a maximum of 6.67 m. This peak is the result of the combined hydraulic pressure of the 800 m³/s flow being obstructed by the weir and subsequently mixing with the tributary inflow in the lower reach.

The temporal progression to a quasi-steady state also reflects the higher volume being processed by the system. The inlet reached a stable depth of 3.552 m at 8.5

hours (30,600 s). Downstream, the outlet exhibited a steep depth increase for the first 13 hours, followed by a transition to a very mild increment after 40 hours. The outlet finally achieved a quasi-steady state of 6.490 m at 41.5 hours (149,400 s). While a marginal rise continued until the end of the 48-hour period, the system reached effective equilibrium at the 41.5-hour mark.

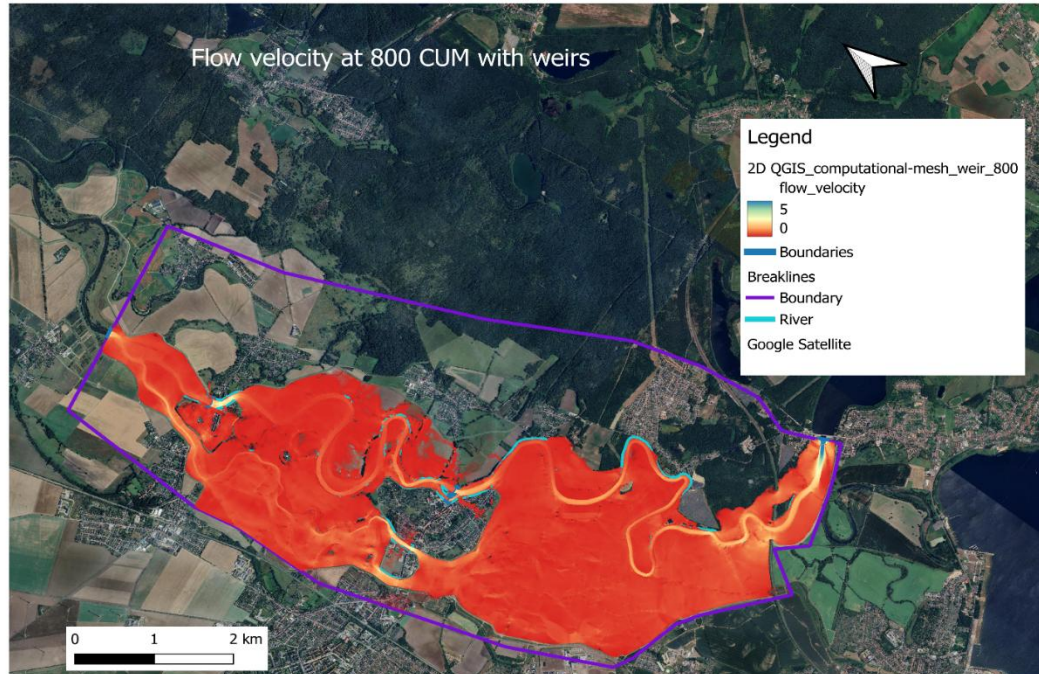


Figure 13: 2D Flow Velocity Map at 800 m³/s discharge with Weir Influence

The increase to 800 m³/s discharge resulted in a global, yet moderate, rise in flow velocities across both the main channel and the floodplains. The maximum velocity observed was 2.5 m/s within the main channel, while floodplain velocities remained relatively low, staying below 0.7 m/s.

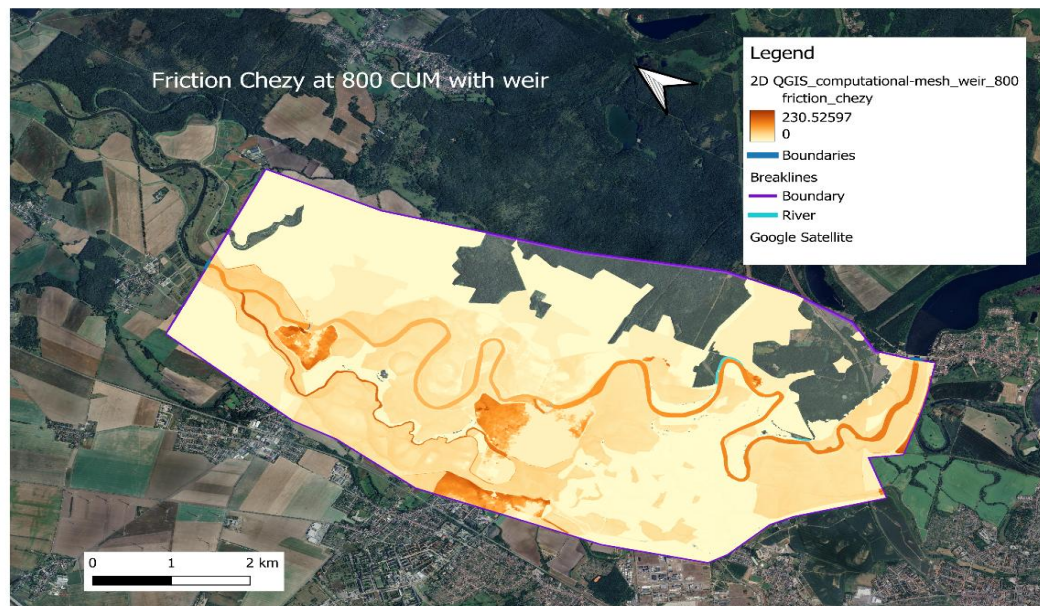


Figure 14: Spatial Distribution of Chézy Friction Coefficients at 800 m³/s with Weir Influence

In this Scenario shown on figure 14, the Chézy friction coefficient was increased to 191 specifically within the inundated sections of the urban zone to account for the increased flow resistance in the flooded city. All other regions remained unchanged from the parameters used in Scenario 1, where these urban friction effects were negligible.

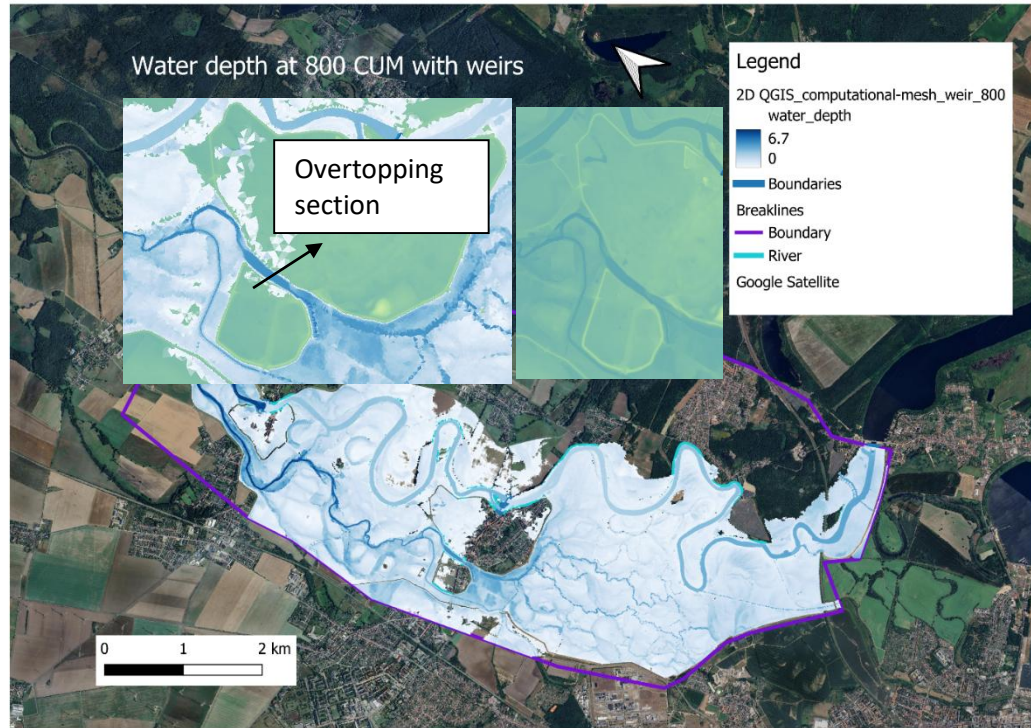


Figure 15: Inundation Depth Map at 800 m³/s with Weir Influence and Identified overtopping section

In this scenario, the simulation results clearly indicate inundation within the urban area. This flooding was not triggered by a structural breach of the dikes; rather, it was caused by elevation deficiencies in the existing embankment system. The Digital Elevation Model (DEM) reveals that the flood protection perimeter (dike) is not a continuous uniform crest elevated loop system in this region.

While the primary dike sections maintain an elevation of approximately 74–75 m, indicated overtopping segments in the figure exhibit a significant elevation deficiency, dropping to 72–73 m. These values are verified from the DEM. During the flood simulation, these low-relief sections acted as preferential inflow paths, allowing water to bypass the primary defenses.

Identifying a continuous 360-degree dike line was challenging due to the resolution limits of the satellite imagery and the DEM. Consequently, these non-continuous depressed segments functioned as hydraulic leakage points. The simulation confirms that in every flooded urban sector, the dike perimeter is interrupted by some topographic depressions at some locations, where the local elevated ground height is insufficient to contain the 800 m³/s flow event.

4. DISCUSSION

The transition from a 1D HEC-RAS model to a 2D BASEMENT simulation revealed critical differences in how hydraulic risks are captured across the Mulde river reach. While the 1D model effectively identified slope-driven velocity spikes, it suffered from significant interpolation limitations. Because HEC-RAS linearly connects water surface elevations between distant cross-sections, the model occasionally caused water to "bleed" over the terrain's bank geometry even in areas where the actual physical channel is narrower. This highlights a lack of lateral precision in the 1D approach, whereas the 2D simulation provided a much more granular view of specific discharge and friction-driven flow. However, the 2D model's accuracy remained heavily dependent on the high-resolution Digital Elevation Model (DEM) and the precision of digitized protective features.

Regarding the Task 2 scenario, the weir crest was set at 73 m, which remains approximately 20 cm below the water surface level under the 500 m³/s discharge condition. Because the elevation difference between the weir section (bed level ≈ 72.4 m) and the adjacent urban terrain (≈ 74 m) is quite small, this height was kept conservative to prevent backwater effects from triggering premature urban flooding. Under these conditions, the flow depth remains essentially unchanged, balancing the need for increased channel capacity with the preservation of upstream urban safety.

When the discharge was increased to 800 m³/s, the hydraulic system's behavior shifted dramatically as the channel capacity was far exceeded. The lateral spread of the flood became much more extensive, pushing water deep into the agricultural and city zones, raising water depths across the entire mesh, and increasing pressure on the identified leakage section. This higher discharge demonstrates that the current dike configurations and conservative weir height are insufficient to protect the urban center under extreme high-flow scenarios in the digital simulation.

However, it is important to note that the simulated urban flooding was primarily caused by elevation deficiencies along the digitized dike lines rather than a hydraulic breach. In a real-world scenario, the urban center could potentially remain protected if temporary flood barriers were deployed at these low-relief inflow points.

Furthermore, certain existing flood protection features may have been omitted from the DEM because their physical dimensions were smaller than the model's grid resolution (e.g., a narrow flood wall). This suggests that while the current simulation identifies vulnerability at these specific locations, the actual risk may be mitigated by physical interventions or existing terrain features that were not captured in the digital elevation data.

5. CONCLUSION AND OUTLOOK

This study utilized 2D hydro-numeric modeling to assess flood propagation and inundation patterns in the Mulde floodplain using QGIS and BASEMENT. The transition from 1D HEC-RAS to a 2D approach provided a more granular view of flow distribution, though it highlighted that model reliability is heavily dependent on the quality of the Digital Elevation Model (DEM) and the precision of digitized protective structures.

The findings indicate that for moderate discharge scenarios ($500 \text{ m}^3/\text{s}$), the flood protection system is generally sufficient, with floodwaters remaining well-contained within the dikes. However, the $800 \text{ m}^3/\text{s}$ simulation exposed significant vulnerabilities, showing an extensive lateral spread of water that penetrated deep into agricultural and city zones as the channel and weir capacity were exceeded.

The simulation further revealed that identified risks to settlements and agricultural lands were directly linked to localized elevation deficiencies in the terrain. In particular, the simulated urban flooding was caused by segments along the dike line where the ground elevation drops to 72–73 m, which is significantly lower than the primary protection height of 74–75 m.

Because of this elevation deficiency, these sections acted as natural inflow points rather than barriers. In a real-world scenario, the deployment of temporary flood protection structures in these specific lower-elevation areas could significantly mitigate the risks shown in the model. These results underscore that flood vulnerability in this region is driven by non-uniform dike crest heights.

To strengthen future flood resilience planning, the current model can be further improved by refining input data, particularly land use classifications, vegetation influences, and sediment transport. Future studies should explore discharge scenarios exceeding $800 \text{ m}^3/\text{s}$ to test the ultimate limits of the existing infrastructure while incorporating climate change projections and potential alterations in river morphology due to erosion and deposition. Overall, this study provides a solid foundation for future improvements, ensuring that flood protection strategies remain effective under evolving hydrological conditions.

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APPENDIX

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